COMP 8347 — Lab #4 Report

Query Section (Part 2.6 and Part 3)

Project: Django 6.0.5, app myapp. All queries reproducible via:

```
python manage.py shell < queries.py
```

Part 2.6 — Basic queries

```
import django from myapp.models import Publisher, Book, Member, Order
```

Screenshot — full shell output (a–d)

```
(.venv) → DjangoProject git:(main) .venv/bin/python manage.py shell -c "
from myapp.models import Publisher, Book, Member, Order
print('=== a. Books ==='); print(Book.objects.all())
print('=== b. Members ==='); print(Member.objects.all())
print('=== c. Orders ==='); print(Order.objects.all())
print('=== d. Publishers ==='); print(Publisher.objects.all())
"
16 objects imported automatically (use -v 2 for details).

=== a. Books ===
<QuerySet [<Book: Machine Learning For Dummies>, <Book: Data Science For Dummies>, <Book: Artificial Intelligence>, <Book: Computer Networking>, <Book: The Night Circus>, <Book: The Underground Railroad>, <Book: Becoming>, <Book: A Walk in the Woods>]>
=== b. Members ===
<QuerySet [<Member: Elena Kwon (elena)>, <Member: Marcus Reed (marcus)>, <Member: Priya Shah (priya)>, <Member: James Bennett (james)>, <Member: Aisha Ncube (aisha)>, <Member: Leo Kwon (leo)>]>
=== c. Orders ===
<QuerySet [<Order: Order #1 - elena (Borrow)>, <Order: Order #2 - marcus (Borrow)>, <Order: Order #3 - aisha (Borrow)>, <Order: Order #4 - james (Borrow)>, <Order: Order #5 - leo (Purchase)>, <Order: Order #6 - elena (Purchase)>, <Order: Order #7 - aisha (Purchase)>]>
=== d. Publishers ===
<QuerySet [<Publisher: Wiley>, <Publisher: Pearson>, <Publisher: Penguin Random House>]>
(.venv) → DjangoProject git:(main)
```

a. List all the books in the db

```
Book.objects.all()
```

Result: Machine Learning For Dummies, Data Science For Dummies, Artificial Intelligence, Computer Networking, The Night Circus, The Underground Railroad, Becoming, A Walk in the Woods

b. List all the members in the db

```
Member.objects.all()
```

Result: Elena Kwon, Marcus Reed, Priya Shah, James Bennett, Aisha Ncube, Leo Kwon

c. List all the orders in the db

```
Order.objects.all()
```

Result: 7 orders — #1 elena Borrow, #2 marcus Borrow, #3 aisha Borrow, #4 james Borrow, #5 leo Purchase, #6 elena Purchase, #7 aisha Purchase

d. List all the publishers in the db

```
Publisher.objects.all()
```

Result: Wiley, Pearson, Penguin Random House

Part 3 — Query practice

Screenshot — full shell output (a–o)

```
(.venv) → DjangoProject git:(main) python manage.py shell < queries.py
16 objects imported automatically (use -v 2 for details).


=== a. Members with last name 'Kwon' ===
<QuerySet [<Member: Elena Kwon (elena)>, <Member: Leo Kwon (leo)>]>

=== b. Publishers in 'USA' ===
<QuerySet [<Publisher: Wiley>, <Publisher: Penguin Random House>]>

=== c. Members living in 'Ottawa' ===
<QuerySet [<Member: Marcus Reed (marcus)>, <Member: James Bennett (james)>, <Member: Leo Kwon (leo)>]>

=== d. Members on an 'Avenue' in ON ===
<QuerySet [<Member: James Bennett (james)>]>

=== e. Members who borrowed 'The Night Circus' ===
<QuerySet [<Member: Elena Kwon (elena)>, <Member: Marcus Reed (marcus)>, <Member: James Bennett (james)>]>

=== f. Books costing more than $40.00 ===
<QuerySet [<Book: Artificial Intelligence>, <Book: Computer Networking>, <Book: The Night Circus>, <Book: Becoming>]>

=== g. Members NOT in province ON ===
<QuerySet [<Member: Elena Kwon (elena)>, <Member: Priya Shah (priya)>, <Member: Aisha Ncube (aisha)>]>

=== h. Orders placed by 'Elena' ===
<QuerySet [<Order: Order #1 - elena (Borrow)>, <Order: Order #6 - elena (Purchase)>]>

=== i. Members with status 'Regular member' ===
<QuerySet [<Member: Marcus Reed (marcus)>, <Member: Priya Shah (priya)>]>

=== j. Books 300-500 pages, category Science&Tech ===
<QuerySet [<Book: Machine Learning For Dummies>, <Book: Data Science For Dummies>]>

=== k. First names of Members who borrowed exactly 2 books ===
<QuerySet ['Elena', 'James', 'Marcus']>

=== l. Books Marcus is borrowing ===
<QuerySet [<Book: Artificial Intelligence>, <Book: The Night Circus>]>

=== m. Members in ON with auto_renew enabled ===
<QuerySet [<Member: Marcus Reed (marcus)>, <Member: James Bennett (james)>]>

=== n. Books Leo purchased ===
<QuerySet [<Book: The Night Circus>, <Book: A Walk in the Woods>]>

=== o. Publisher city of books purchased by Elena ===
<QuerySet ['London']>
(.venv) → DjangoProject git:(main)
```

a. Members whose last name is 'Kwon'

```
Member.objects.filter(last_name='Kwon')
```

Result: Elena Kwon, Leo Kwon

b. Publishers with headquarters in 'USA'

```
Publisher.objects.filter(country='USA')
```

Result: Wiley, Penguin Random House

c. Members that live in 'Ottawa'

```
Member.objects.filter(city='Ottawa')
```

Result: Marcus Reed, James Bennett, Leo Kwon

d. Members that live on an 'Avenue' and live in ON province

```
Member.objects.filter(address__icontains='Avenue', province='ON')
```

Result: James Bennett

Elena also lives on an Avenue but in BC, so correctly excluded.

e. Members that have borrowed the book 'The Night Circus'

```
Member.objects.filter(borrowed_books__title='The Night Circus')
```

Result: Elena Kwon, Marcus Reed, James Bennett

f. Books that cost more than $40.00

```
Book.objects.filter(price__gt=40.00)
```

Result: Artificial Intelligence ($197.32), Computer Networking ($143.99), The Night Circus ($41.00), Becoming ($45.00)

g. Members that do NOT live in province ON

```
Member.objects.exclude(province='ON')
```

Result: Elena Kwon (BC), Priya Shah (SK), Aisha Ncube (MB)

h. Orders placed by a client whose first_name is 'Elena'

```
Order.objects.filter(member__first_name='Elena')
```

Result: Order #1 (Borrow), Order #6 (Purchase)

i. Members whose status are 'Regular Member'

```
Member.objects.filter(status=1)
```

Result: Marcus Reed, Priya Shah

j. Books with 300–500 pages (inclusive) in category 'Science&Tech'

```
Book.objects.filter(num_pages__range=(300, 500), category='S')
```

Result: Machine Learning For Dummies (464p), Data Science For Dummies (432p)

k. First name of Members who have borrowed exactly 2 books

```
from django.db.models import Count
Member.objects.annotate(n=Count('borrowed_books')).filter(n=2).values_list('first
_name', flat=True)
```

Result: Elena, James, Marcus

l. Books that Member with username 'Marcus' is currently borrowing

```
Book.objects.filter(member__username='marcus')
```

Result: Artificial Intelligence, The Night Circus

m. Members who live in ON and have auto_renew enabled

```
Member.objects.filter(province='ON', auto_renew=True)
```

Result: Marcus Reed, James Bennett

n. Books that 'Leo' has purchased

```
Book.objects.filter(order__member__username='leo', order__order_type=0)
```

Result: The Night Circus, A Walk in the Woods

order_type 0 = Purchase.

o. City where the headquarters of the publisher of the book purchased by 'Elena' is located

```
Publisher.objects.filter(     books__order__member__first_name='Elena',
books__order__order_type=0, ).values_list('city', flat=True).distinct()
```

Result: London

Elena purchased 'Artificial Intelligence' → published by Pearson → headquartered in London.